

HIGH- PERFORMANCE ALGORITHMS FOR TOMOGRAPHY

Multi-scale correction of optical diffusion in X-ray tomography

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Central question

How can we correct the diffusion halo without paying the computational cost of the full correction?

Where does the diffusion come from?

From X-rays to the measured projection



$$I_{\text{meas}} = I_{\text{ideal}} * K$$

I_{meas} : measured detector projection
 I_{ideal} : ideal projection before optical diffusion
 K : spatial diffusion kernel
 $*$: convolution operator

Light generated in the scintillator spreads before reaching the detector.

The diffusion is multi-scale

Several characteristic spatial lengths coexist in the same kernel

$$K(r) \propto \delta(r) + \sum_i f_i \frac{e^{-r/s_i}}{r + \varepsilon}$$

r : distance from the kernel center

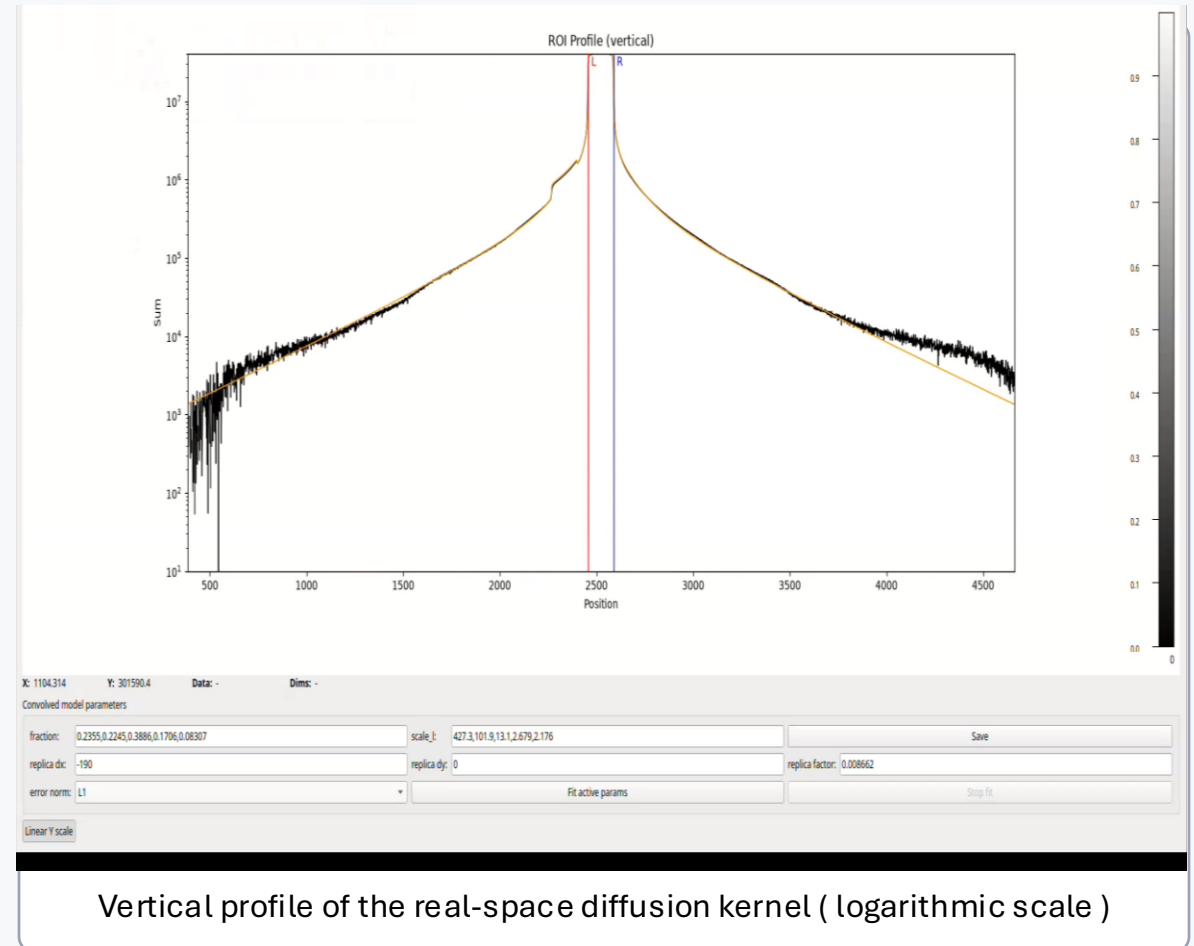
s_i : characteristic diffusion length of component i

f_i : weight / fraction of component i

$\delta(r)$: direct, non-diffused contribution

Short range → local variations

Long range → extended halo



Why is the full correction expensive?

Reference approach: full-resolution deconvolution

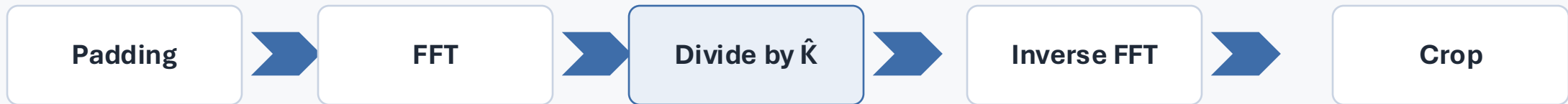
$$\hat{I}_{\text{corr}} = \frac{\hat{I}_{\text{meas}}}{\hat{K}}$$

$\hat{I}$: Fourier transform

I_{corr} : corrected projection

I_{meas} : measured projection

$\hat{K}$: diffusion kernel in Fourier space



The full correction processes large projections at full spatial resolution, even though part of the correction varies only slowly.

Separate low and high spatial frequencies

The correction is decomposed into a smooth global part and a detailed local part

$$Q = I_{\text{meas}} - I_{\text{corr}}$$

$$Q = Q_{\text{LF}} + Q_{\text{HF}}$$

$$Q_{\text{LF}} = G_{\sigma} * Q$$

$$Q_{\text{HF}} = Q - Q_{\text{LF}}$$

Q: diffusion correction

Q_LF: low-frequency part

Q_HF: high-frequency part

G_{σ} : Gaussian filter

σ : Gaussian width

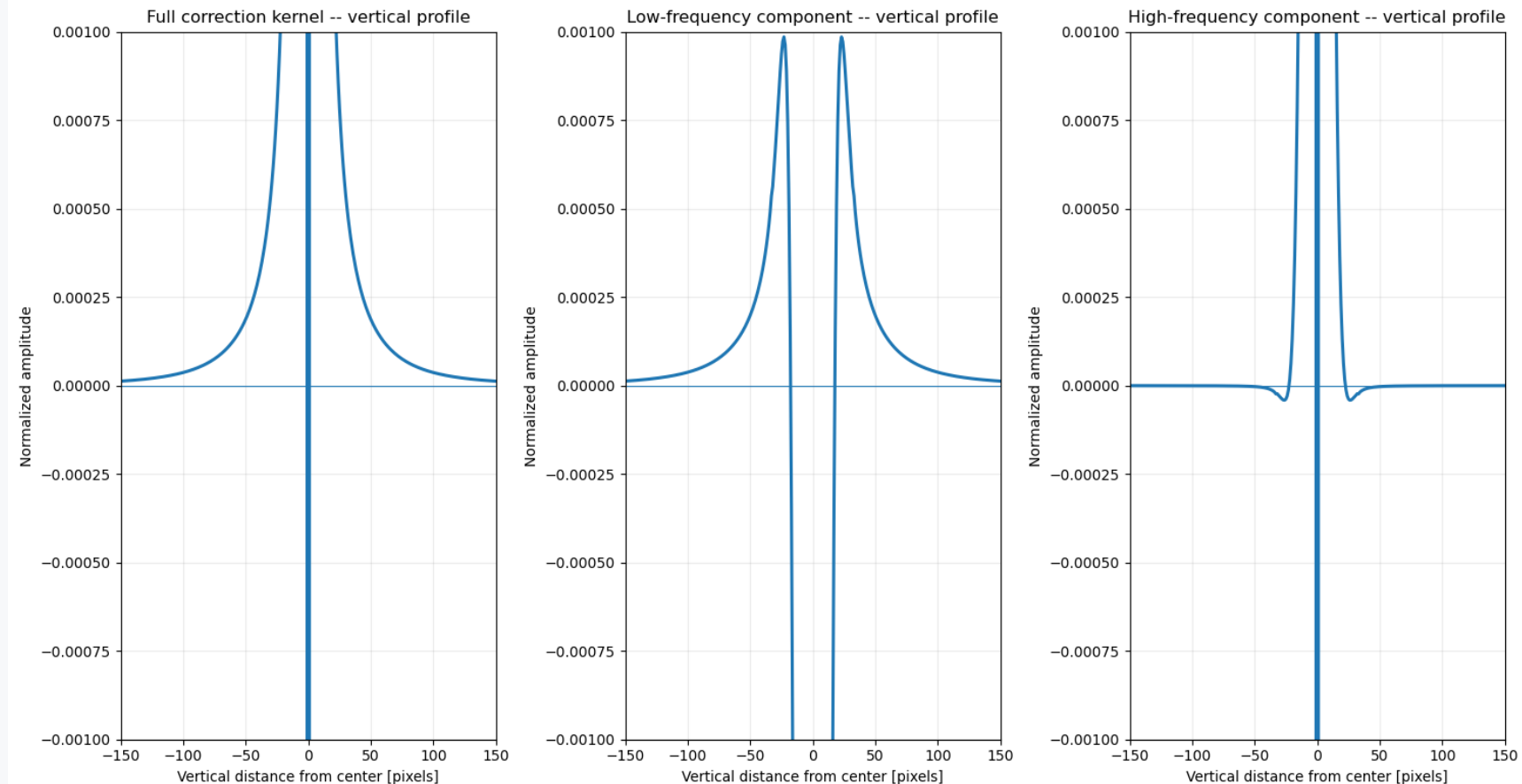
*: convolution

$\sigma = 8$ pixels

LF: smooth + extended

HF: detailed + localized

Correction kernel separation -- Gaussian filter, sigma = 8



Correction kernel decomposition into low- and high-frequency components using a Gaussian filter ($\sigma=8$). The low-frequency component captures the spatially extended contribution, while the high-frequency component retains the localized structures.

Why can the low-frequency part be cheaper?

A smooth field can be represented with fewer spatial samples

$$Q_{\text{LF}}^{(b)} = \text{bin}_b(Q_{\text{LF}})$$

b: spatial binning factor

N_x, N_y : original image dimensions

spatial bin = 4

$$N_x N_y \rightarrow \frac{N_x N_y}{b^2} \quad (b = 4)$$

≈ 16× fewer spatial samples

Second observation :

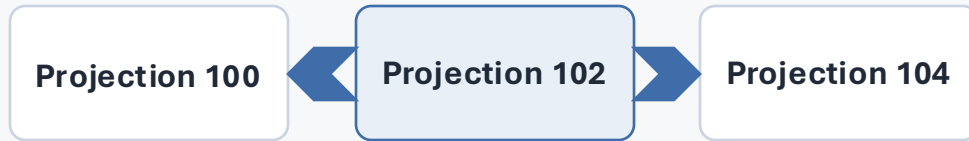
The low-frequency correction also changes slowly between nearby projections.

→ This suggests reducing the angular sampling as well.

Data reduction ≠ measured end-to-end speed-up. A final speed-up requires benchmarking the complete implementation.

We do not need the LF correction for every projection

Predict projection 102 from two nearby corrected projections



$$Q_{LF}(102) \approx \left[(1 - t) \frac{Q_{LF}(100)}{C_{100}} + t \frac{Q_{LF}(104)}{C_{104}} \right] C_{102}$$

$t = (102 - 100) / (104 - 100) = 1/2$: interpolation weight

C_i : beam-current normalization for projection i

$Q_{LF}(i)$: LF correction of projection i

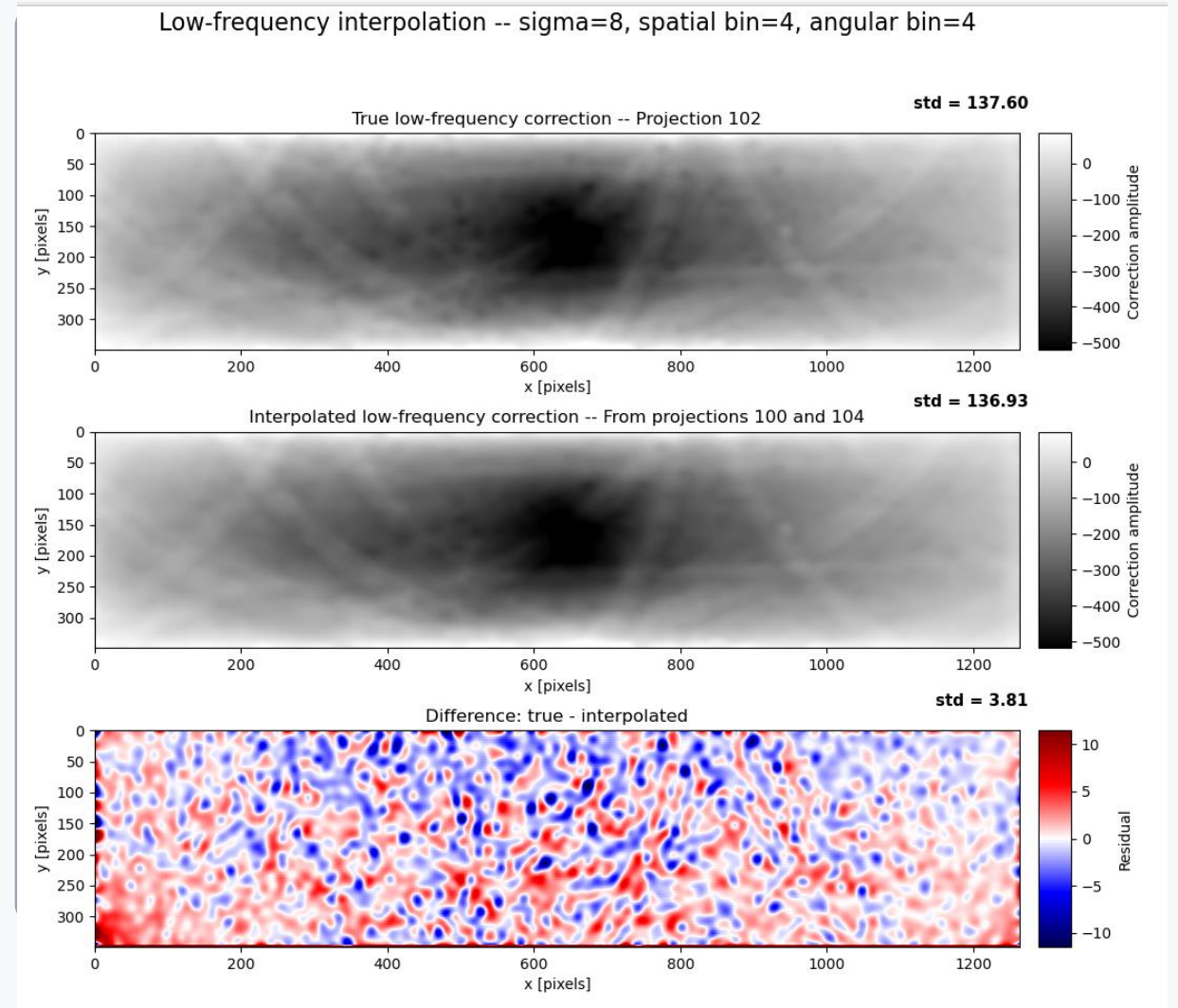
Residual fluctuations are small compared with the LF correction :

$\text{std}(\text{true LF}) = 137.6$

$\text{std}(\text{interpolated LF}) = 136.9$

$\text{std}(\text{residual}) = 3.81$

→ residual std $\approx 2.8\%$ of true-LF std



Low-frequency correction at projection 102, predicted from projections 100 and 104 using spatial binning $\times 4$ and angular interpolation $\times 4$. The residual (right) is displayed with a symmetric diverging scale centered at zero.

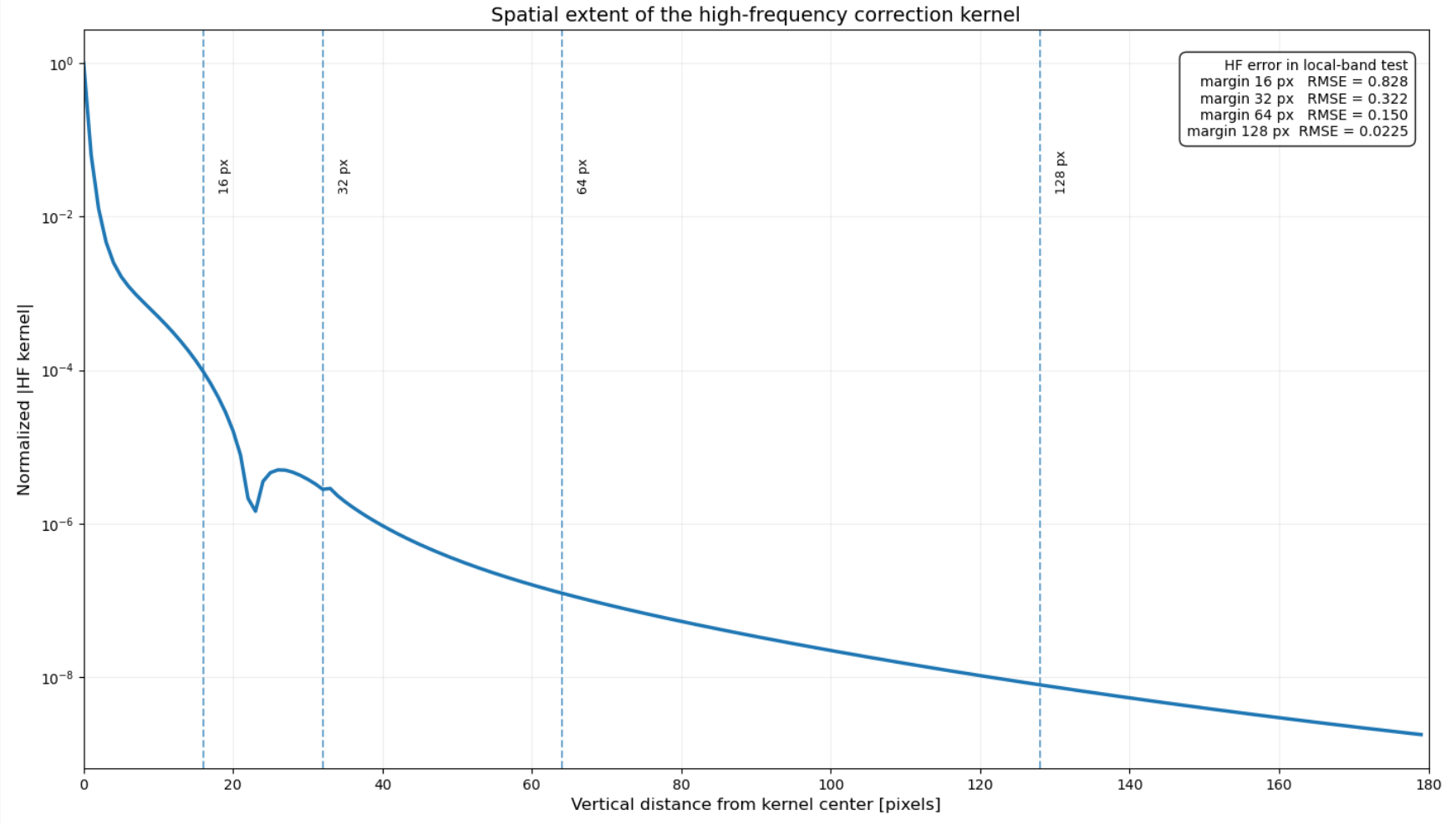
The high-frequency part is local

Fine spatial details must be preserved, but they do not require the full projection height

$$Q_{\text{HF}} = Q - Q_{\text{LF}}$$

Q_{HF} : local high-frequency correction
 Q : total correction
 Q_{LF} : smooth low-frequency correction

**HF amplitude
decreases rapidly with
distance from the
kernel center.**



Normalized vertical HF-kernel profile. Tested margins: 16, 32, 64, 128 px. HF RMSE: 0.828 → 0.322 → 0.150 → 0.0225.

Combine both approximations

Global low frequency + local high frequency

$$Q_{\text{approx}} = Q_{\text{LF, pred}} + Q_{\text{HF, local}}$$

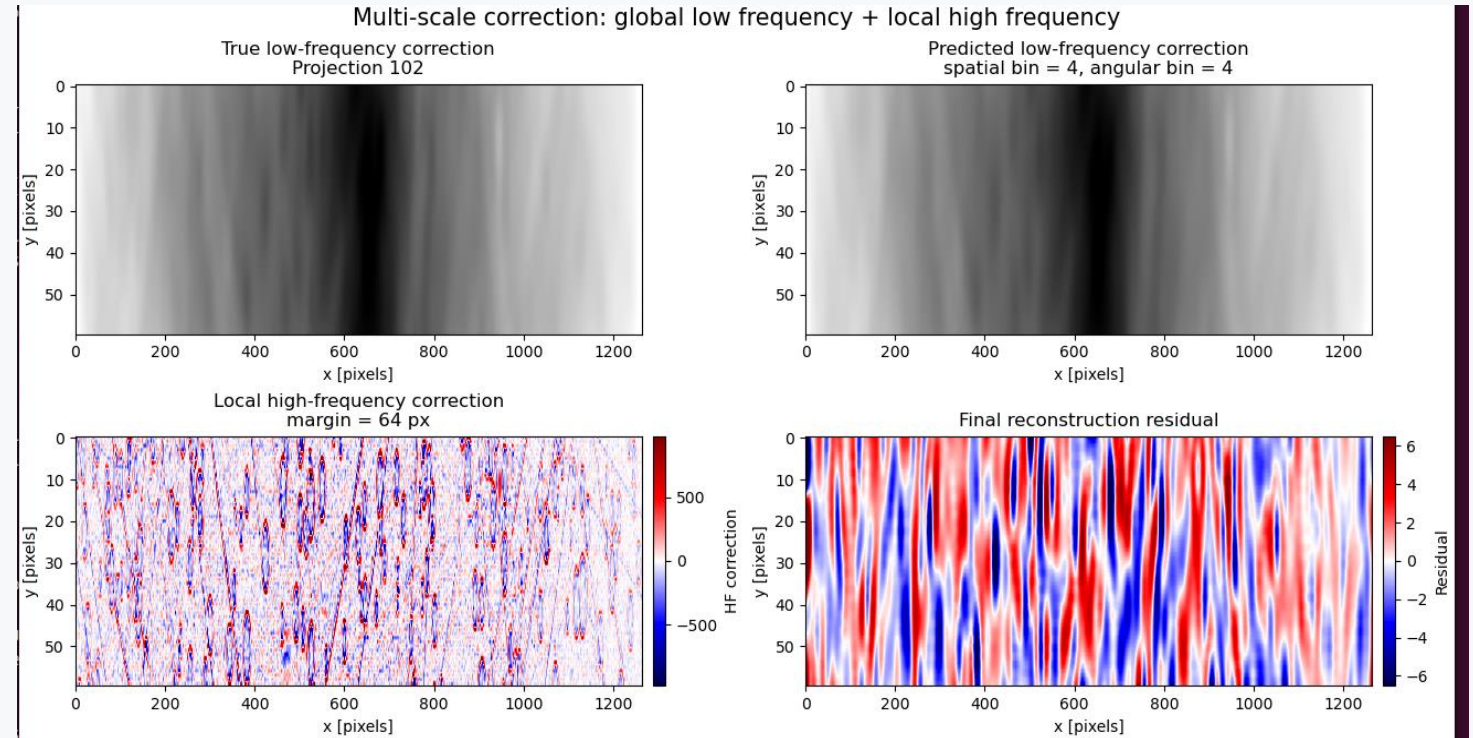
$$I_{\text{corr, approx}} = I_{\text{meas}} - Q_{\text{approx}}$$

$Q_{\text{LF, pred}}$: binned + angularly interpolated LF term

$Q_{\text{HF, local}}$: HF term computed on the local band

$I_{\text{corr, approx}}$: final approximate corrected projection

- $\sigma = 8$
- spatial bin = 4
- angular bin = 4
- HF margin = 64 px

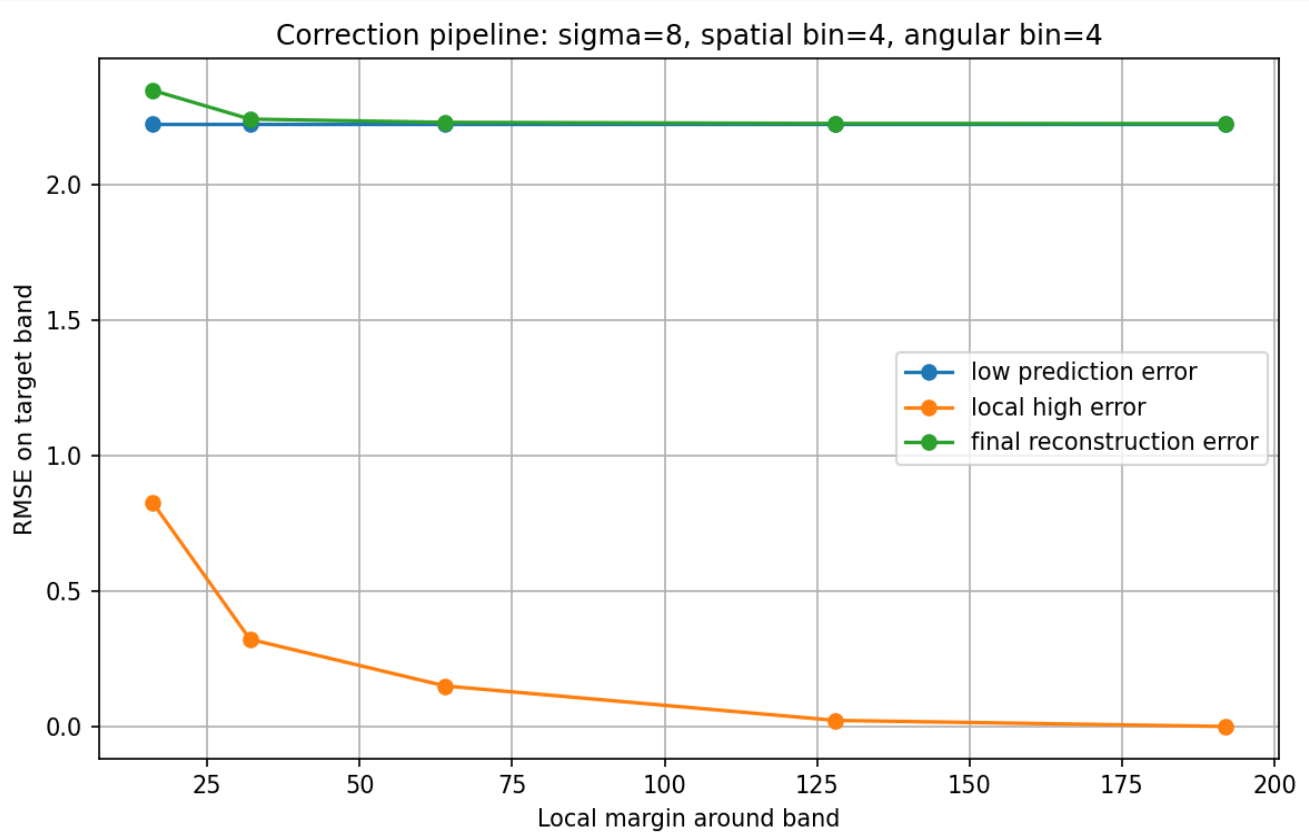


Combined multi-scale correction: spatially and angularly interpolated low-frequency contribution combined with a local high-frequency correction using a 64-pixel margin. The final residual is shown relative to the full-reference correction.

Where is the remaining error?

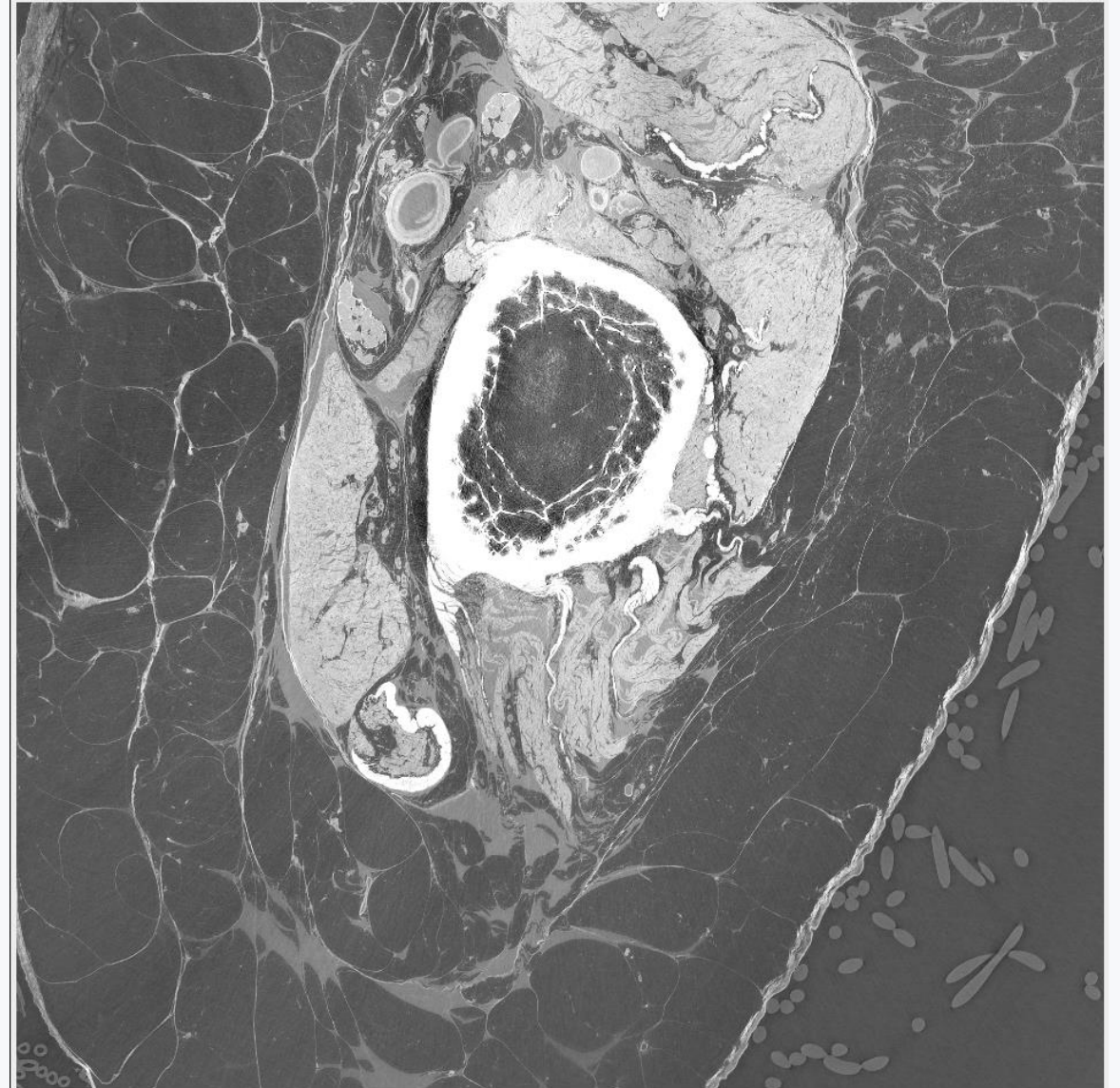
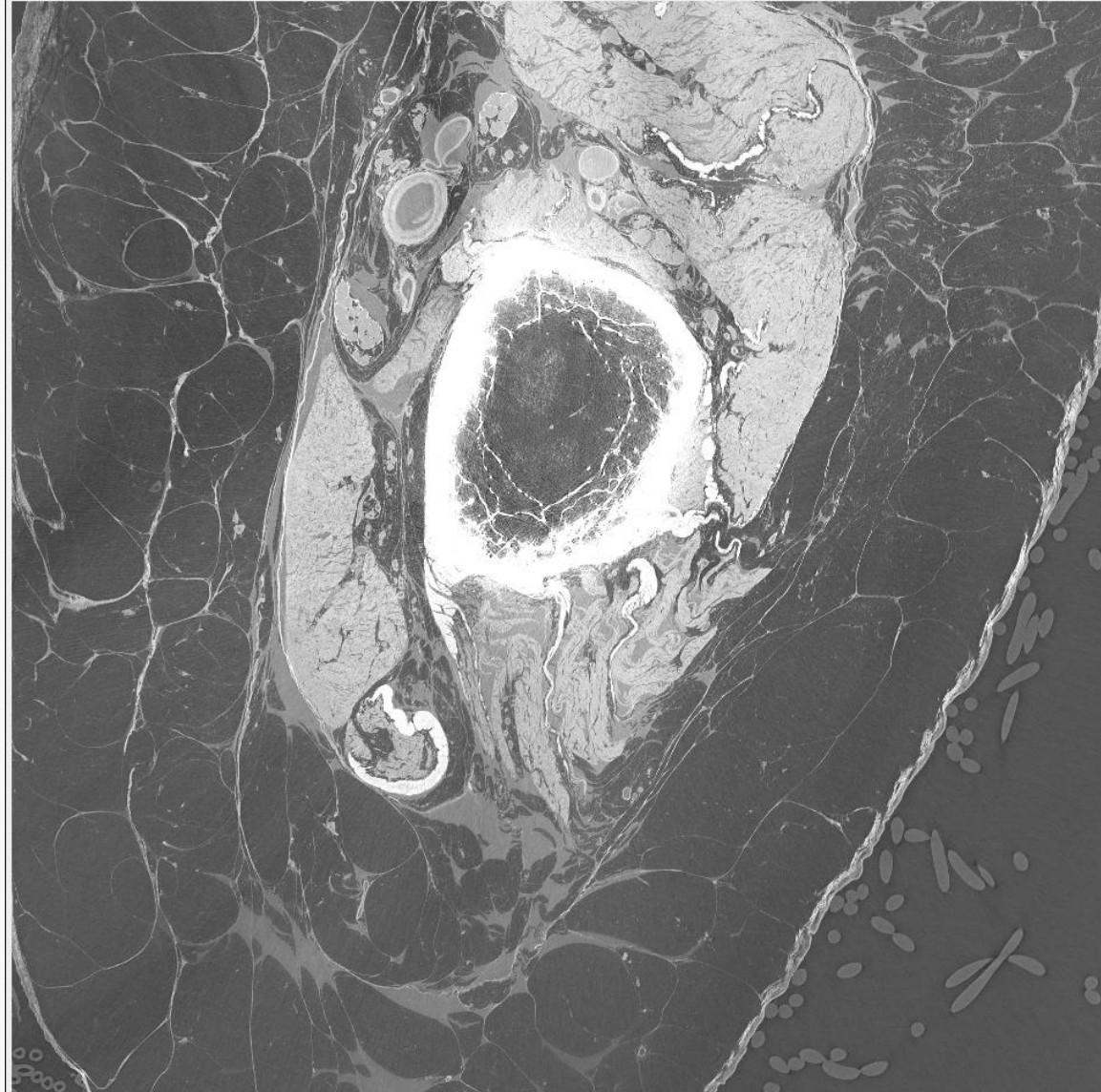
Once HF is sufficiently localised, the bottleneck shifts to the LF approximation

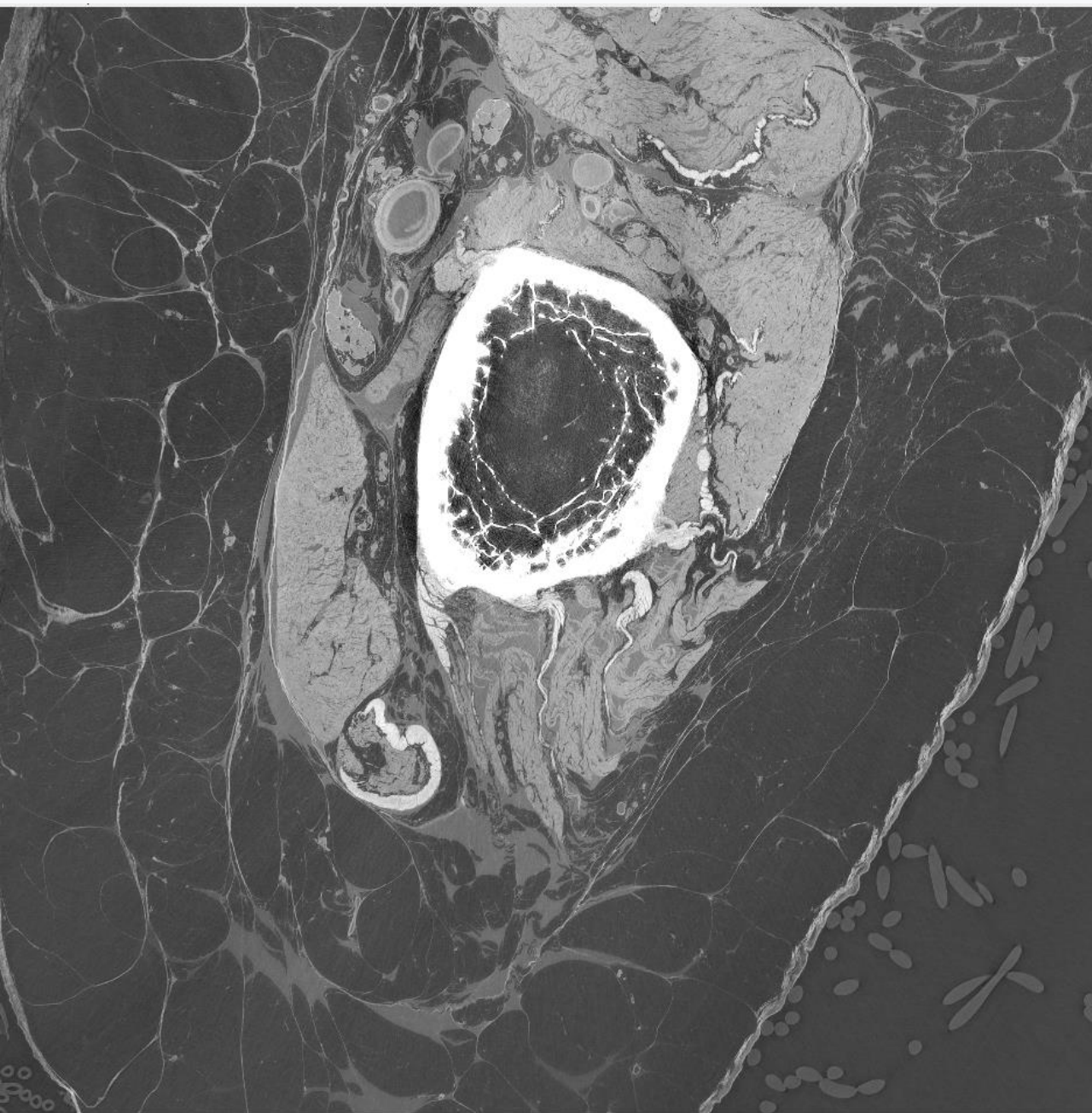
HF margin	HF RMSE	Final RMSE
16 px	0.828	2.349
32 px	0.322	2.243
64 px	0.150	2.231
128 px	0.0225	2.226
192 px	0	2.226

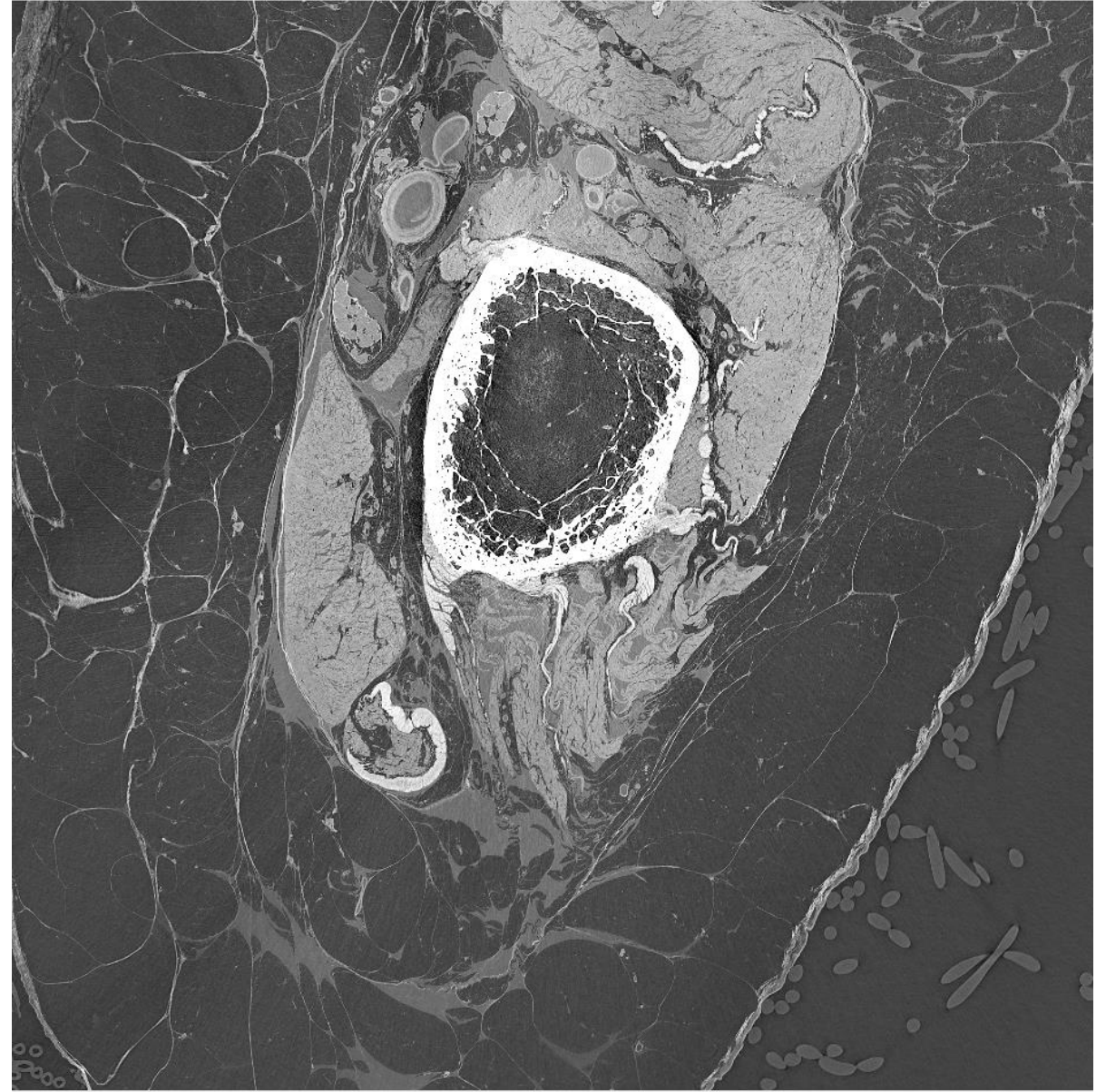


As the HF error approaches zero, the final RMSE saturates near the LF prediction error (~2.23).

Beyond ~64–128 px, increasing the HF margin brings little improvement to the final error.







Measured performance gains

DISK FOOTPRINT

4.7 TB → 80 GB

≈ 59× smaller

98.3% reduction

DIGEST CREATION

32 min → 4 min

8× faster

87.5% less time

RECONSTRUCTION

118 min → 93 min

–25 min

≈ 21.2% less time

Measured final gains

$4.7 \text{ TB} / 80 \text{ GB} \approx 59$

$32 / 4 = 8$

$(118 - 93) / 118 \times 100 \approx 21.2\%$

Comments after presentation :

- Model is Linear : separation of L_f and H_f .
- Add pipeline recap : steps $\Rightarrow$ diff from previous pipeline
- Diffusion + problem of correlation of neighbor pixels
- Diffusion comes from SENSOR !!!!
- Optical ghosting : LENS !!!
- From high resolution x rays (low wavelengths) $\Rightarrow$ visible light : we introduce some degradation
- Why diffusion and what exactly it is